YASASWI BHUMIREDDY

AI/ML ENGINEER

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LinkedIn: linkedin.com/in/yasaswireddy | GitHub: github.com/yasaswireddy7032

Portfolio: yasaswireddy7032.github.io/evidence-search/

SUMMARY

AI/ML engineer with 4+ years of experience across financial services and cloud AI consulting. Builds generative AI workflows, retrieval pipelines, and Python services with a focus on evaluation, latency, and reliability. Current work at Citibank supports risk and compliance investigations.

PROFESSIONAL EXPERIENCE

AI Engineer | Citibank, Risk & Compliance Technology

February 2025 - Present | New York, NY

- Built a Vertex AI/Gemini Pro compliance system for transaction review, KYC screening, and SAR drafting, reducing case time from 34 to 13 minutes and increasing analyst throughput 2.6x.
- Orchestrated LangGraph and LangChain workflows for alert triage, sanctions screening, and entity resolution, with deterministic Python risk scoring; reduced investigation latency 47%.
- Built hybrid retrieval and reranking across 14K+ policy pages using LlamaIndex, OpenSearch, and Hugging Face embeddings, achieving 94% retrieval precision.
- Implemented PII redaction, output filtering, and source-grounding checks, reducing false regulatory citations from 8.2% to 0.6%.
- Served inference through FastAPI with bounded concurrency, deadlines, retry budgets, canary releases, and rollback, sustaining 99.9% availability.

Delivery Consultant - Gen AI/ML | Amazon Web Services

July 2024 - January 2025 | Arlington, VA

- Built an AI self-service platform using Bedrock and Connect, processing 2.1M monthly conversations and reducing repeat contacts 18%.
- Benchmarked conversational and tool-using workflows; used prompt caching, batching, and context compaction to reduce serving costs 28%.
- Built retrieval and tool-calling workflows with LangChain and LlamaIndex. Prepared 68K labeled cases with PySpark for Nova fine-tuning, improving accuracy from 84% to 92%.
- Used staged releases, automated rollback, and CloudWatch, X-Ray, and OpenTelemetry monitoring to reduce triage time 41%.

Machine Learning Engineer | Jio Payments Bank

March 2021 - July 2023 | Hyderabad, India

- Built Kafka and Flink pipelines processing 35K card and UPI events per second, with PySpark transformations for training and real-time fraud features.
- Trained XGBoost and LightGBM models with graph-based entity features, improving fraud recall 14% and reducing false positives 19%.
- Built FastAPI scoring services with AsyncIO and Redis. Automated tracking and releases with MLflow, Airflow, Docker, and Kubernetes, cutting release lead time from 3 weeks to 4 days.

ENGINEERING PROJECTS & BACKGROUND

SELECTED INDEPENDENT PROJECTS

Cited Research Agent | Python, Ollama, retrieval, evaluation

github.com/yasaswireddy7032/cited-research-agent

- Implemented a bounded agent loop that selects search and document-read actions, checks exact source quotes, fingerprints cited documents, and supports abstention.
- Added nine passing tests covering unread citations, fabricated quotes, rejected tools, loop limits, and the mocked model adapter contract.

Approval Triage Agent | Python, SQLite, agent workflows

github.com/yasaswireddy7032/approval-triage-agent

- Built policy-guided routing proposals with persistent review state. Human confirmation is tied to the exact payload before a local assignment is written.
- Added nine passing tests for approval matching, persistence, changed proposals, and idempotent execution after restart.

Evidence Search | Python, FastAPI, SQLite, BM25

github.com/yasaswireddy7032/evidence-search

- Built tenant-scoped retrieval with reader/writer authorization, SQLite persistence, versioned citations, and integration tests for access isolation and audit privacy.

Projects use demonstration data and are separate from employment work. New agents include scripted demos and optional Ollama integration; live-model behavior remains to be evaluated.

TECHNICAL SKILLS

Languages and services: Python, SQL, Java, TypeScript, FastAPI, AsyncIO, REST, GraphQL

Generative AI: Vertex AI, Gemini, Amazon Bedrock, PyTorch, Hugging Face, LangGraph, LangChain, LlamaIndex, retrieval-augmented generation, tool calling, model evaluation

Retrieval and data: OpenSearch, FAISS, Pinecone, PostgreSQL, Redis, Kafka, Flink, PySpark, Airflow, BigQuery

Deployment and operations: AWS, Google Cloud, Docker, Kubernetes, Terraform, CI/CD, MLflow, OpenTelemetry, Prometheus, Grafana, CloudWatch, X-Ray

EDUCATION

M.S. in Data Science & Business Analytics | University of North Carolina at Charlotte

August 2023 - December 2024 | Charlotte, NC | GPA: 3.7

CERTIFICATIONS

AWS Certified Generative AI Developer - Professional (AIP-C01) | Amazon Web Services

Meta Back-End Developer Professional Certificate | Coursera

Meta Front-End Developer Professional Certificate | Coursera

RESEARCH PUBLICATIONS

Melanoma Classification Using Deep Transfer Learning. IJDIIC, 1(1), 11-20, 2022. DOI: 10.59461/ijdiic.v1i1.8

BOARD #322: An NSF FMRG Supported Exploratory Study of Prompting Large Language Models for a Conversational Manufacturing Education Platform. ASEE Annual Conference, 2025. DOI: 10.18260/1-2--55689